

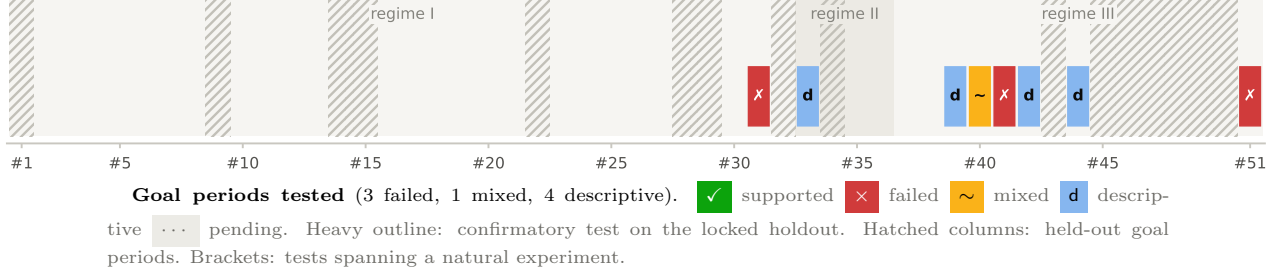
H78: Replicator growth order: herding vs division of labor

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1/5

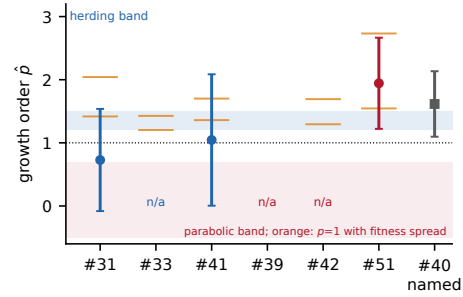
Bottom line. Repo growth order is not identifiable from commit logs: first-order copying plus fitness spread explains every $\hat{p}$, and herding sits on kickoff-named repos.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

H78 treats each repo as a replicator and fits its growth order p on the swarm call clock. Herding weeks should give $p = 1.2\text{--}1.5$ and own-artifact weeks $p \leq 0.7$. The data are DQ4 work commits in three herding weeks (#31, #33, #41), three own-artifact points (#39, #42, #51) and two natives (#40, #44). Field-made arrivals are removed: births, kickoff-named repos (H54) and the H28 blind window. The herding weeks give $\hat{p} = 0.73 [-0.08, 1.54]$ and $1.05 [0.00, 2.09]$. #51 gives $1.94 [1.22, 2.67]$ from 271 recruitments, which is the wrong sign. First-order copying ($p = 1$) with a spread of repo fitness reproduces every value. Formation makes 76–100% of arrivals. The holdout is not run.



Observables. The growth order $\hat{p}$ per period (95% CI, formation-free recruitments, host expiry $E = 100$). The blue band is the herding prediction and the red band the parabolic prediction. Orange bars show the median $\hat{p}$ of simulated worlds on the same call schedule with $p = 1$ and repo fitness spread $\sigma_A = 0.5$ and 1.0 . Every testable estimate lies near a $p = 1$ world. The gray square is the #40 hub, a kickoff-named repo, which also grows superlinearly.

PHYSICAL MODEL

Model 05 (replicators) with model 06 as the null. A replicator is a repo j . Its copy number n_j is the count of agents whose current work label is j . A recruitment moves an agent's label into a repo that already has hosts. C_j^{free} counts the calls of agents who do not host j . κ_j is formation by a field (births, kickoff naming). A_j is a repo's fitness (attractiveness). The clock τ counts swarm calls (H40). $p = 1$ is neutral copying, $p > 1$ conformist, $p < 1$ parabolic.

$$\frac{dn_j^+}{d\tau} = \kappa_j + c A_j \frac{C_j^{\text{free}}}{C} n_j^p$$

$$R_{jb} \sim \text{Pois}(C_{jb}^{\text{free}} e^{\alpha + p \log n_{jb}}), \quad n_{jb} \geq 1$$

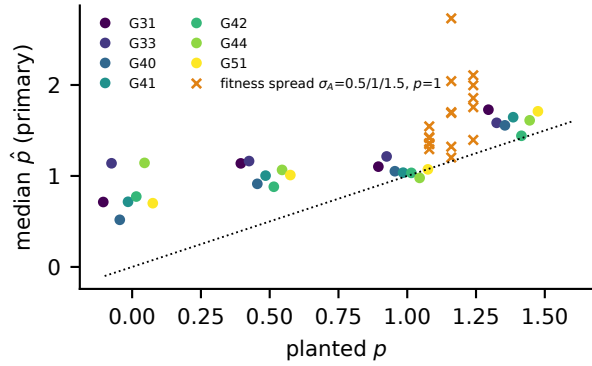
$$A_j \sim \text{LogN}(0, \sigma_A^2) \Rightarrow \hat{p} > p \text{ (pooled over repos)}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A monitor cannot read “winner-take-all dynamics” from how fast big repos recruit. Fitness differences between repos produce the same superlinear signal. Herding in these weeks is a repo that the kickoff names, and it takes every arrival. To prevent duplicated effort, an operator should control what the kickoff names. Watching the recruitment law does not help. The blind window is negligible in commit data (0.7% of recruitments). Most recruitments (66%) are returns to a repo the agent worked on before. This mostly restates H54 and H58 in replicator terms.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\hat{p} \in [1.2, 1.5]$, herding	#31 0.73, #41 1.05; #33 9 events	failed
P2 $\hat{p} \leq 0.7$, own	#51 1.94 [1.22, 2.67]; #39, #42 untestable	failed
P3 $\Delta p > 0$	-1.05	failed
P4 impostors shift < 0.2	#31 0.81, #33 0.33, #41 0.05	failed
P5 cross-lab within 0.3	#31 0.42, #41 -0.72, #51 2.43	failed
P7 recovery $p \in \{0.5, 1, 1.5\}$	$p = 1, 1.4$ recovered; 0.5 reads ≈ 1.0	failed
N #40 named hub $\hat{p} \leq 0.5$	1.62 [1.10, 2.13]; all 48 into the hub	mixed
A0 step from the field	calibrated in 2/6 (#41, #51); burst 3.9×	failed



Second figure. Synthetic recovery on the real schedules, 40 runs per world (10 for #51), measured through the same label builder. Planted $p = 1$ and $p = 1.4$ are recovered. A planted $p = 0.5$ and a field-only world ($p = 0$) read near 1. Worlds with $p = 1$ and repo fitness spread (orange) read 1.2–2.7.

GOAL PERIODS IN DETAIL

G31 × failed
 G33 d descriptive
 G39 d descriptive
 G40 ~ mixed
 G41 × failed
 G42 d descriptive
 G44 d descriptive
 G51 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The host expiry E sets which losses count. $E = 300$ inflated $\hat{p}$ through idle “ghost” hosts, so $E = 100$ was fixed before the real run (A1). The H54 naming rule tags 46–100% of recruitments. Agents name repos after the goal, so “named” is partly agent-made. The fitness worlds, the touch-based impostor class and the AR(1) step null were added after the first real run (A2). #33, #42 and #44 have under 15 formation-free recruitments. Commits are attempts.

ROUND 2

Within-repo dynamic-panel estimators validated on the fitness worlds; repo covariates as fitness; an explicit return term.

- **H78-R1** Separate order from fitness.
- **H78-R2** Model returns explicitly.
- **H78-R3** A better naming instrument.